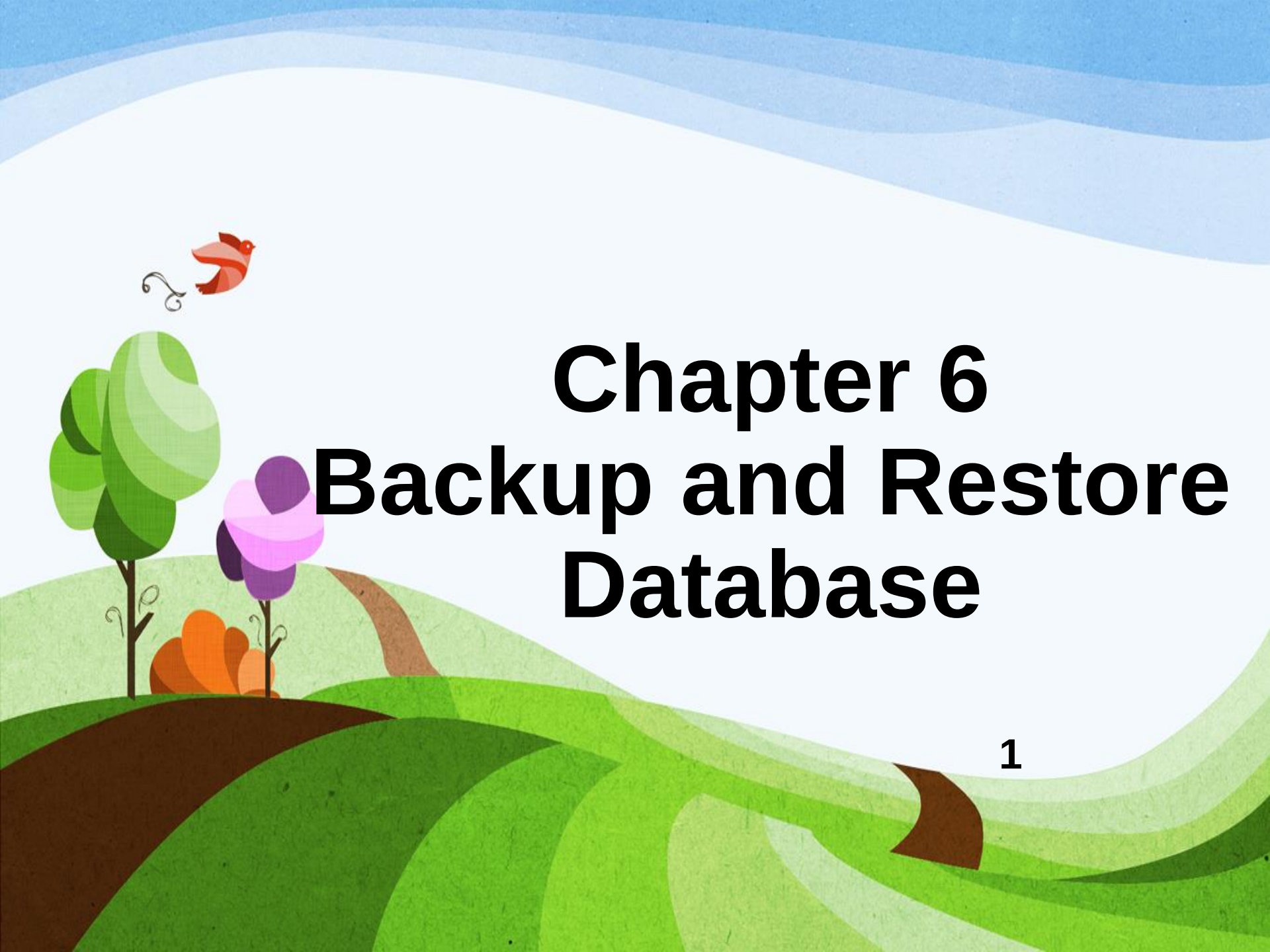


A stylized, colorful illustration of a landscape. In the foreground, there are rolling green hills with a dark brown path winding through them. On the left, there are three trees: a tall green one, a purple one, and an orange one. A small red bird is flying in the sky above the trees. The background consists of layered, wavy blue and white shapes representing a sky or distant hills.

Chapter 6

Backup and Restore

Database

Chapter Outline

- Introduction to Database Backup
- Database Recovery Models
- Database Backup Types
- Restore Databases

Introduction to Database Backup

- Along with security, one of the fundamental task of a database administrator is to ensure that data can be recovered in the event of a disaster.
- Having a database backup strategy, standby servers, and multiple copies of a database are all best practices for database management.
- The first task to be performed before taking the backup of the database is setting up the **recovery model**.

Database Recovery Models

- **Recovery model** is a database property setting that determines the types of database backups for recovering the database.
- The recovery models available in SQL Server are:
 - **Full recovery:** all database changes are logged to the transaction log.
 - **Bulk-logged:** all changes are logged except those that cause large log entries.
 - **Simple recovery:** database changes are not logged to the transaction log. This method can be used for a database that is never changed.

Database Backup Types

- Basic database backup types available in SQL Server for each recovery model are given below:

Recovery Model	Backup Type		
	Full	Differential	Transaction Log
Full	Yes	Yes	Yes
Bulk-logged	Yes	Yes	Yes with minimal logged
Simple	Yes	Yes	No

Full Database Backup

- A full database backup backs up the whole database.
- It is the starting point for every backup procedure and recovery process.
- For a small database, the best practice is to use just full database backup.
- However, as a database becomes larger, full backup takes more time to finish and requires more storage space. Therefore, for a large database, you might want to supplement full database backup with differential backup.

Differential Database Backups

- A differential backup captures all pages that have changed since the last full backup, even if there has been a subsequent differential backup.
- A differential backup has to be applied to a full backup and can't exist until a full backup has been created.
- At restore time, the full backup is restored first, followed by the most recent differential backup.

Transaction Log Backups

- Contain all the changes that have occurred since the last transaction log backup.
- Have any transactions that were created after a full or differential backup.
- **Tail log transaction log** backup captures all transactions since the last transaction log backup.
- The first operation that should be performed during a corrupted database restore process is to take a final transaction log backup.
 - This step in the restore process is referred to as backing up the **tail of the log**.

Restoring Databases

- Every database restore begins with a full backup followed by a differential and transaction log backups.
- Depending on the types of backup, you may restore:
 - Full backups only
 - Full and differential backups
 - Full and transaction log backups
 - Full, differential, and transaction log backups